

Financial Frictions, Used Capital, and Misallocation

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Abstract. Standard macro-finance models treat capital as homogeneous. Financially constrained firms can therefore adjust only the quantity of capital they operate, not its vintage. This assumption is at odds with the data: young, small, and likely constrained firms use second-hand equipment disproportionately. I build a dynamic macro-finance model in which firms rent both new and used capital. Second-hand capital lets constrained firms operate closer to their efficient scale, even when they cannot meet the collateral requirements for new capital. It thereby reduces the dispersion of marginal revenue products of capital and weakens the dependence of entrepreneurship on initial wealth. I calibrate the model to Vietnamese firms and find that used capital mitigates the allocative and occupational distortions created by financial frictions.

JEL codes: E22, E44, L26, O47.

Keywords: Financial frictions, used capital, misallocation, entrepreneurship, aggregate productivity.

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I. Introduction

Why is productivity so much lower in poor countries? Differences in physical and human capital explain only part of the large gaps in output per worker across countries. Much of the gap is attributed to differences in total factor productivity. One explanation is that resources do not always reach the producers who can use them most productively. Policies, institutions, and market frictions can keep productive firms from expanding, reducing aggregate productivity. Misallocation can therefore help explain why productivity is lower in developing countries ([Restuccia and Rogerson, 2008](#); [Hsieh and Klenow, 2009](#); [Restuccia and Rogerson, 2017](#)).

Capital misallocation is one source of these productivity losses ([David and Venkateswaran, 2019](#); [Bau and Matray, 2023](#)). Financial frictions can prevent productive firms with little wealth from obtaining the capital they need. A large quantitative literature studies this mechanism in dynamic general equilibrium models with collateral constraints ([Buera et al., 2011](#); [Moll, 2014](#); [Midrigan and Xu, 2014](#)). A common feature of these models is that capital is homogeneous: a financially constrained firm can adjust the quantity of capital it operates, but not whether that capital is new or used. This omits an adjustment margin that matters in the data, since young and financially constrained firms rely disproportionately on used equipment ([Eisfeldt and Rampini, 2007](#); [Ma et al., 2022](#)). How much does access to used capital change the aggregate productivity losses caused by financial frictions?

I address this question by introducing a choice between new and used capital into a dynamic general equilibrium model with collateral constraints and endogenous occupational choice. Firms rent both new and used capital from a financial intermediary, choosing the mix jointly with their operating scale. The model therefore links firms' use of used equipment to their productivity and financial position. It also allows second-hand capital to affect both the allocation of capital across operating firms and the decision to become an entrepreneur.

The central mechanism is a trade-off between the rental cost of used equipment and the collateral required to rent it. Used equipment may carry a higher rental cost per physical unit. Still, because its market value is lower, it absorbs less of a constrained entrepreneur's financing capacity per dollar of rental spending. A productive entrepreneur with little wealth may consequently rent used capital and operate at a scale that new capital alone would not finance. As wealth rises, the collateral constraint becomes less important and firms' choices converge toward the unconstrained mix. The same mechanism raises the value of entrepreneurship for productive but poor agents and thus weakens the dependence of occupational choice

on initial wealth.

I quantify this mechanism using firm-level data on Vietnamese small and medium-sized enterprises. These data offer a rare opportunity to study this margin: the Vietnam SME Survey is a firm-level panel that follows the same firms across six rounds from 2005 to 2015 and records, in every round, the share of each firm’s main equipment acquired second-hand, alongside its assets, debt, and access to credit (CIEM et al., 2005–2015). The benchmark economy contains both new and used capital; the comparison economy removes second-hand capital. I calibrate the two separately, so that each reproduces the same five targets, including the debt-to-output ratio and the top-decile employment share; the benchmark additionally matches the decline in used-capital reliance with firm assets and the aggregate ratio of new to used capital. I then remove the collateral constraint within each calibrated economy and recompute equilibrium prices, occupational choices, and stationary distributions. The comparison identifies both the contribution of second-hand capital and the cost of the collateral constraint.

Quantitatively, used capital has two effects. First, it improves how capital is allocated: relative to a single-vintage economy – one in which firms rent only new capital, calibrated on its own to the same five empirical targets – a second-hand capital reduces the misallocation that collateral constraints generate by 39 percent. Second, it changes who runs a firm. Collateral constraints keep many talented but poor agents out of entrepreneurship, and a second-hand capital repairs 23 percent of that lost entry, while entry by talented rich agents is nearly unchanged. Entrepreneurship therefore depends much less on initial wealth.

This paper contributes to the literature on factor misallocation and financial frictions, in which a constrained firm can adjust only the quantity of capital it operates.² This paper adds the choice of which kind of capital to operate, and that choice turns out to matter: access to second-hand capital removes a substantial part of the misallocation that collateral constraints generate in an economy without a second-hand capital. The paper also relates to research on entrepreneurship and wealth.³ I differ from that work in giving the constrained agent a second capital type rather than only a smaller scale, and I show that this changes who becomes an entrepreneur: because used equipment absorbs less financing capacity per dollar of rental spending, a talented agent can enter on less wealth than the collateral constraint

²On the measurement of misallocation and its aggregate cost, see Restuccia and Rogerson (2008), Hsieh and Klenow (2009, 2014), David and Venkateswaran (2019), Bau and Matray (2023), and Adamopoulos and Restuccia (2014). On financial frictions in dynamic general equilibrium models with collateral constraints, see Buera et al. (2011), Moll (2014), Midrigan and Xu (2014), and Chen et al. (2023); Buera et al. (2015).

³Evans and Jovanovic (1989), Cagetti and De Nardi (2006), Buera and Shin (2013), and Tan (2025).

would otherwise require.

The rest of the paper is organized as follows. Section II relates the analysis to the literatures on misallocation, financial frictions, and used capital. Section III presents the motivating evidence from the Vietnamese SME survey. Section IV develops the model. Section V presents the calibration and quantitative results. Section VI concludes.

II. Related Literature

This paper builds on four related literatures: misallocation, financial frictions, entrepreneurship and wealth, and used capital. It examines how firms' choice between new and used equipment changes the productivity losses and occupational decisions caused by collateral constraints.

Misallocation and its measurement. Restuccia and Rogerson (2008) show that distortions across heterogeneous producers can generate large aggregate productivity losses. Hsieh and Klenow (2009) quantify the potential gains from reallocating capital and labor across manufacturing plants in China and India. Restuccia and Rogerson (2017) survey this literature and emphasize the importance of identifying the specific frictions behind misallocation.

David and Venkateswaran (2019) distinguish several sources of dispersion in average revenue products of capital among Chinese manufacturers. Adjustment costs and uncertainty together explain only a small part of this dispersion; most is associated with productivity-correlated or permanent firm-specific factors. Financial frictions and size-dependent policies can contribute to these patterns, but the decomposition does not identify a unique underlying cause. Hsieh and Klenow (2014) document that manufacturing plants in India and Mexico grow much less with age than US plants. Their models imply that these differences in life-cycle growth could substantially lower aggregate manufacturing productivity. These findings motivate studying specific frictions that limit firm growth and distort resource allocation.

Distortions also affect who becomes a producer. Using farm-level panel data from China, Adamopoulos et al. (2022) show that distortions affect both resource allocation across farmers and selection into farming. The selection response amplifies aggregate productivity losses. My paper studies both effects in a model with collateral constraints: the allocation of capital across operating firms and the occupational choices of productive people with little wealth. I examine how access to used equipment changes these outcomes.

Financial frictions in quantitative general equilibrium. A large quantitative

literature studies financial frictions in models with heterogeneous producers. [Buera et al. \(2011\)](#) show that financial frictions distort the allocation of capital and entrepreneurial talent, with larger effects in sectors that require a larger scale of operation. Saving can partly offset these constraints. [Moll \(2014\)](#) shows that the effectiveness of self-financing depends on the persistence of productivity: sufficiently persistent shocks can imply small steady-state losses but slow transitions. [Buera and Shin \(2013\)](#) show that financial frictions slow resource reallocation after reforms, generating gradual increases in investment and productivity. [Midrigan and Xu \(2014\)](#) find that capital misallocation among existing producers generates relatively small productivity losses, while distorted entry and technology adoption account for larger losses. [Buera et al. \(2015\)](#) survey this literature.

[Chen et al. \(2023\)](#) show that managerial inputs can weaken the self-financing mechanism. As firms grow, their expenditure share on managers rises, reducing profit margins and slowing wealth accumulation. Managerial inputs also increase capital demand. Together, these effects make financial constraints more persistent, including among large firms, and amplify productivity losses. My paper adds a choice between new and used equipment. Alongside saving and adjusting their scale, firms can change their capital mix in response to a collateral constraint. I quantify how this choice changes the constraint's aggregate effects.

Entrepreneurship and wealth. The relationship between wealth and entrepreneurship is central to models of occupational choice with borrowing constraints. [Evans and Jovanovic \(1989\)](#) study how liquidity constraints affect entrepreneurial selection. [Cagetti and De Nardi \(2006\)](#) show how borrowing constraints and entrepreneurial saving jointly shape business ownership and the wealth distribution. [Fonseca et al. \(2007\)](#) show in a model that start-up costs can weaken the relationship between wealth and entrepreneurship and provide supporting cross-country evidence from older adults.

[Tan \(2025\)](#) studies how capital resale frictions affect entrepreneurial investment and the wealth distribution. When capital is illiquid, entrepreneurs face greater liquidity risk and lower collateral values, which discourages investment and worsens capital misallocation. His mechanism runs through the difficulty of reselling entrepreneurial capital: limited resale options raise the cost of adjusting investment and weaken the collateral backing of entrepreneurial assets, with consequences for both aggregate efficiency and the wealth distribution.

[Guntin and Kochen \(2026\)](#) study how trade in existing firms can alleviate financial constraints. They document that many US entrepreneurs purchased their business rather than founding it, and that buyers are wealthier than sellers on average. Following

a sale, capital grows faster than output, reducing the firm’s average revenue product of capital. Their model shows how transferring ownership can improve capital allocation.

My paper connects occupational choice to firms’ choice between new and used equipment. In my model, used equipment absorbs less financing capacity per dollar of rental spending. Access to used capital can therefore make entrepreneurship more attractive to productive people with little wealth. I quantify how this choice changes the dependence of entrepreneurship on wealth.

Used capital and financing. [Eisfeldt and Rampini \(2007\)](#) study why financially constrained firms use second-hand capital. They show that the share of used capital in capital expenditures declines with firm size and is much higher among the smallest firms than among the largest. Used-capital investment is also associated with measures of financial constraints. Their mechanism works through the timing of payments: used equipment requires a smaller initial payment but higher subsequent maintenance costs, making it attractive to firms that are currently constrained. [Ma et al. \(2022\)](#) document a related pattern: young firms purchase disproportionately old equipment, with evidence consistent with financial constraints.

The financing advantages of different assets also depend on their durability and collateral value. [Rampini and Viswanathan \(2013\)](#) study how collateral constraints affect investment, financing, and leasing. [Rampini \(2019\)](#) shows that more durable assets require larger down payments because durability raises their price by more than their collateral value. Financially constrained firms therefore prefer less durable and used assets. My model incorporates a choice between new and used capital into a general equilibrium setting where firms rent both types from a financial intermediary and agents choose between entrepreneurship and wage work.

Related work examines used-capital prices and resource allocation. [Lanteri \(2018\)](#) documents that capital reallocation and used-capital prices are procyclical. His model explains these patterns through firm-level capital specificity, which makes new and used investment goods imperfect substitutes. My model also treats new and used capital as imperfect substitutes. [Edgerton \(2011\)](#) estimates the elasticity of substitution between new and used equipment for farm and construction machinery and finds that the two are substitutes, but imperfect ones, providing an empirical benchmark for this parameter.

[Lanteri and Rampini \(2023\)](#) study the efficiency of capital reallocation when firms face collateral constraints. Used-capital prices affect both collateral values and the allocation of resources between buyers and sellers. [Bierdel et al. \(2023\)](#) study how asymmetric information about capital quality distorts trade in illiquid markets, with

larger effects during crises. My paper examines how access to used capital changes the productivity losses and occupational choices caused by collateral constraints. I quantify these effects using Vietnamese firm-level panel data that combine information on second-hand equipment use with firms' financial positions.

III. Evidence on Used Capital and Financing Frictions

A. The Survey

I use the Vietnamese Small and Medium Enterprise Survey (CIEM et al., 2005–2015), a biennial firm-level panel of small and medium-sized manufacturing enterprises run by CIEM, ILSSA, the University of Copenhagen, and UNU-WIDER. Owners or managers are interviewed in person in ten provinces, and the same firms are revisited every two years. The six rounds from 2005 to 2015 cover 5,131 firms and 15,811 firm-wave observations.

For each firm the questionnaire records the age of its main equipment, the share acquired new, second-hand, or self-constructed, whether it is owned or rented, and how it was financed, alongside total assets, formal and informal debt, loan applications, whether the firm had problems obtaining a loan, and, for non-applicants, why not.

Three features make the survey suited to the question. The firms are small, mostly household-owned, and largely outside formal credit, which is the population the model describes. Used equipment is common rather than marginal. I can observe a firm's reliance on used equipment change with its own financial position.

B. Used Capital and Financing Frictions

Table 1 reports the used-capital share by firm age. Second-hand equipment accounts for 31.5 percent of a typical young firm's main equipment, against 21.6 percent for firms past eleven years. Similar gradients appear in richer economies (Eisfeldt and Rampini, 2007; Ma et al., 2022). A cross-section cannot tell whether the gradient reflects financing constraints or permanent firm differences. Proposition 2 predicts that within a firm, the used share should fall as collateral accumulates; Section V.A shows μ falling with wealth in the model.

I estimate

$$\text{UsedShare}_{it} = \beta_1 \text{ZeroDebt}_{it} + \beta_2 (\text{ZeroDebt}_{it} \times X_{it}) + \beta_3 X_{it} + \mathbf{\Gamma}' \mathbf{Z}_{it} + \lambda_t + \alpha_i + \varepsilon_{it}, \quad (1)$$

where UsedShare_{it} is used capital as a percentage of firm i 's main equipment and $\mathbf{Z}_{it}$ holds two employment-size indicators. Column (1) sets X_{it} to centered log assets; column (2) sets X_{it} to a young-firm indicator and additionally controls for log assets.

Firm and year fixed effects absorb permanent firm characteristics and common round shocks; identification comes from within-firm changes in debt status, assets, and age. Zero debt marks firms that finance themselves without borrowing, not the levered constrained firms in the model. Because some zero-debt firms simply choose not to borrow, β_2 is likely too small rather than too large. Footnote 4 confirms the pattern using direct reports of credit problems.⁴

Table 1. Used-capital share by firm age

Firm age	Share (%)	Observations
Young (0–5 years)	31.5	2,447
6–10 years	26.0	3,854
11 years or more	21.6	8,610

Notes: Firm-weighted averages over 14,911 firm-wave observations with a usable start year. Author’s calculations from the pooled 2005–2015 waves of the Vietnam SME Survey (CIEM et al., 2005–2015).

Table 2. Financing frictions and used capital

<i>Dependent variable:</i> used-capital share (percentage points)		
	(1) Asset interaction	(2) Young-firm interaction
Zero debt $\times$ log assets	–1.067** (0.462)	—
Zero debt $\times$ young	—	3.885* (2.340)
Zero debt	–0.482 (0.836)	–1.046 (0.872)
Log assets	–0.260 (0.493)	–0.679 (0.453)
Young (age ≤ 5)	—	0.755 (1.658)
Observations	13,492	13,403
Firms	3,551	3,549
Within R^2	0.036	0.036
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Size controls	Yes	Yes

Notes: Standard errors clustered by firm in parentheses. Log assets is centered at the sample mean. Size controls are indicators for 10–49 and 50 or more workers. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The estimates match the collateral prediction. Among zero-debt firms, the used share falls by 1.067 percentage points per log point of assets (column 1) and is 3.885

⁴Firms that applied for a formal loan report whether they had problems obtaining it. In 2013, applicants reporting problems hold 11.7 pp more used capital (s.e. 3.10, $N = 644$); pooling 2011–2015 gives 6.8 pp (s.e. 1.82, $N = 2,010$). Results are similar with controls. Robustness checks on column (2): raw reported age gives 5.345 (2.378); firms in ≥ 3 rounds give 4.801 (2.439).

points higher while the firm is young and zero-debt ($0.755 + 3.885$ in column (2)). The effects are modest relative to a mean used share of 24.5 percent, as fixed effects remove level differences between firms; the point is the sign and the within-firm pattern. The evidence is descriptive rather than causal.

Three points to note. First, there is no unconditional within-firm age gradient. In column (2), *Young* (0.755, s.e. 1.658) is the effect of being young for firms *with debt*; for zero-debt firms the within-firm premium is 4.6 percentage points ($p = 0.027$). The result is therefore the interaction with financing, not age alone. Second, zero debt marks self-financing rather than collateral rationing: most non-applicants say they did not need or want credit (2.6 percent cite inadequate collateral). Footnote 4 checks the pattern among loan applicants who report binding constraints. Third, attrition and the stock (not flow) nature of the vintage item work against finding an effect and align the regressions with the model’s value share rather than purchase flows in Eisfeldt and Rampini (2007); Ma et al. (2022). Taken together, reliance on second-hand equipment falls with a firm’s own assets among plausibly constrained firms.

IV. Model

I develop a stationary general equilibrium model with endogenous occupational choice and a competitive rental market for new and used capital. Agents save through deposits with a competitive financial intermediary that owns the physical capital stock and rents capital services to entrepreneurs. All agents, including the intermediary, are price takers, and all rental rates and asset prices are equilibrium objects. The model departs from a standard collateral-constraint economy in one respect: firms rent both new and used capital, and the collateral constraint is written on the combined market value of the two. Throughout, capital of type $j \in \{n, u\}$ has price p_j and user cost R_j per dollar of its market value, so the rental payment per physical unit is $p_j R_j$.

Notation. Each symbol below has a single meaning throughout the paper. $G(a, z, y)$ is the stationary distribution of agents; the entry shortfall of poor talented agents is written $\mathcal{S}$ (Section V.C). L is labor in efficiency units; the TFP loss from the collateral constraint is written $\mathcal{M}$ (Section V.B). D is aggregate deposits held with the intermediary; a worker’s labor income is written $I_w(y)$, and aggregate external finance is written B^{ext} . Roman letters (G, L, D) denote model objects; calligraphic letters ($\mathcal{M}, \mathcal{S}$) denote the two loss measures used to summarize the quantitative results, and $\mathcal{E}, \mathcal{W}$ denote the sets of entrepreneurs and workers.

A. Environment and Timing

Time is discrete and there is no aggregate uncertainty. A unit measure of infinitely lived agents is indexed by the individual state (a, z, y) , where a is financial wealth, z is log entrepreneurial productivity, and y is worker ability. Preferences are

$$U_0 = E_0 \sum_{t=0}^{\infty} \beta^t u(c_t), \quad u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad (2)$$

with discount factor $\beta \in (0, 1)$ and risk aversion σ .

B. Technology and Idiosyncratic States

Entrepreneurial technology. An entrepreneur with productivity z hires L efficiency units of labor and rents new and used capital, K_n and K_u , to produce

$$Y = \exp(z) (L^\alpha K^{1-\alpha})^\eta, \quad \eta < 1, \quad (3)$$

where α is the labor share within the variable-input bundle and $\eta < 1$ governs the span of control, as in [Lucas \(1978\)](#), so that optimal firm scale is finite and increasing in productivity. New and used capital are imperfect substitutes, with effective capital given by the constant-elasticity aggregate

$$K = \left[\phi^{1/\gamma} K_n^{(\gamma-1)/\gamma} + (1-\phi)^{1/\gamma} K_u^{(\gamma-1)/\gamma} \right]^{\gamma/(\gamma-1)}, \quad (4)$$

where $\gamma > 1$ is the elasticity of substitution between new and used capital and ϕ their relative weight. Entrepreneurial productivity is persistent:

$$z' = \begin{cases} z, & \text{with probability } \omega, \\ \tilde{z}', \quad \tilde{z}' \sim F_z, & \text{with probability } 1 - \omega, \end{cases} \quad (5)$$

so ω governs the persistence of productivity. The induced distribution of productivity levels, $\exp(z)$, has a Pareto right tail with parameter η_p .

Workers. A worker with ability y supplies y efficiency units inelastically and earns labor income

$$I_w(y) = Wy, \quad (6)$$

where W is the wage per efficiency unit. Ability follows

$$\log y' = \rho_y \log y + \sigma_y \varepsilon'_y, \quad \varepsilon'_y \sim N(0, 1), \quad (7)$$

with stationary standard deviation $\sigma_y/\sqrt{1-\rho_y^2}$. Productivity and ability evolve independently.

C. Individual Decisions and Occupational Choice

Entrepreneurial profits. Conditional on operating a firm, an agent with wealth a and productivity z chooses labor and both new and used capital to maximize operating profits,

$$\pi(a, z) = \max_{K_n, K_u, L \geq 0} \left\{ \exp(z) \left(L^\alpha K^{1-\alpha} \right)^\eta - WL - p_n R_n K_n - p_u R_u K_u \right\}, \quad (8)$$

subject to K in (4) and the financing constraint

$$p_n K_n + p_u K_u \leq \theta a, \quad (9)$$

where a is the entrepreneur's asset holdings used as collateral and $\theta \geq 1$ governs the stringency of the collateral constraint, with a smaller θ indicating a tighter constraint. New and used capital each absorb financing capacity equal to their own market value.

Consumption and saving. The value conditional on entrepreneurship is

$$V_e(a, z, y) = \max_{\substack{c \geq 0 \\ a' \geq 0}} \{u(c) + \beta E[\Omega(a', z', y') \mid z, y]\}, \quad (10)$$

subject to

$$c + a' = \pi(a, z) + (1 + r)a, \quad (11)$$

and the value conditional on wage work is

$$V_w(a, z, y) = \max_{\substack{c \geq 0 \\ a' \geq 0}} \{u(c) + \beta E[\Omega(a', z', y') \mid z, y]\}, \quad (12)$$

subject to

$$c + a' = I_w(y) + (1 + r)a. \quad (13)$$

Deposits are non-negative: agents cannot borrow against future labor income, and the only credit in the economy is the intermediated capital limited by (9).

Occupational choice. The value available at (a, z, y) is

$$\Omega(a, z, y) = \max \{V_e(a, z, y), V_w(a, z, y)\}, \quad (14)$$

and the agent operates a firm if

$$V_e(a, z, y) \geq V_w(a, z, y) \quad (15)$$

and supplies labor otherwise. These occupational flows, together with the resulting composition of the entrepreneurial sector, are equilibrium outcomes of (15).

D. The Financial Intermediary and Capital Prices

A competitive financial intermediary collects deposits at the risk-free rate r , owns the physical capital stock, and rents capital services to entrepreneurs. It earns zero profit. The final good is the numeraire and one unit of it becomes one unit of new capital, so $p_n = 1$ throughout. The stocks evolve as

$$K_{n,t+1}^s = (1 - \delta - \zeta)K_{n,t}^s + I_t, \quad (16)$$

$$K_{u,t+1}^s = (1 - \delta - \kappa)K_{u,t}^s + \zeta K_{n,t}^s, \quad (17)$$

where I_t is investment, δ is baseline depreciation, $\zeta \in (0, 1)$ is the fraction of new capital that becomes used each period, and κ is the fraction of used capital that breaks and is hence valueless, with $\delta + \zeta < 1$ and $\delta + \kappa < 1$. Used capital is formed from new capital that becomes used, so it is in positive supply without absorbing investment of its own. Write p_u for the price of used capital relative to new, so that $p_u R_u$ is the rental payment per physical unit.

Proposition 1 (Capital prices). *Let the intermediary discount at r , take prices as given, and choose investment to maximize the present value of rental income net of investment subject to (16)–(17), where new capital rents at $p_n R_n$ and used capital at $p_u R_u$ per physical unit. Then a stationary solution with $K_n^s, K_u^s \in (0, \infty)$ requires*

$$R_n = r + \delta + \zeta \left(1 - \frac{p_u R_u}{r + \delta + \kappa} \right). \quad (18)$$

The proof is in Appendix A.

Only the product $p_u R_u$ appears in (18), and only that product enters a firm's rental bill, so the model determines the rental payment on used capital rather than its two factors. Define

$$R_u = r + \delta + \kappa, \quad (19)$$

the user cost of used capital: one dollar held in used capital must earn the deposit rate, cover depreciation, and cover the rate at which the machine breaks. With R_u so

defined, (18) determines the price of used capital relative to new,

$$p_u = p_n - \frac{R_n - r - \delta}{\zeta}, \quad (20)$$

and (18) can be written $R_n = r + \delta + \zeta(p_n - p_u)$. The last term is the net cost of aging: with probability ζ a unit priced at p_n is repriced at p_u . At $\zeta = 0$ it collapses to $R_n = r + \delta$, which is the case in the economy without used capital.

This split would be immaterial if used capital merely produced. It matters here because the financing constraint (9) is written on market value, which makes p_u enter the entrepreneur's problem on its own. The mechanism of the paper originates in it: breakage raises R_u through (19), making used capital expensive to rent, while the price at which its rental stream capitalizes is determined separately by (20), making it cheap in collateral capacity.

Deposits are the intermediary's only liability and capital its only asset, so

$$D = p_n K_n^s + p_u K_u^s. \quad (21)$$

Given (21), the pricing condition implies zero profit, as Appendix A shows. The intermediary is a veil: the economy behaves as if households held the capital stock directly and priced it at (p_n, p_u) .

In a stationary equilibrium the used stock is constant, so

$$\zeta K_n^s = (\delta + \kappa) K_u^s, \quad \text{hence} \quad \frac{K_u^s}{K_n^s} = \frac{\zeta}{\delta + \kappa}. \quad (22)$$

This ratio is fixed by (ζ, δ, κ) and does not respond to prices; the levels of both stocks scale with aggregate wealth, as (27) makes explicit. Three prices clear three markets: the wage clears the labor market, the deposit rate clears the new-capital market through (21), and R_n adjusts until the demand ratio matches (22). By (20) a higher R_n lowers p_u , which tilts demand toward used capital through both the rental payment and the collateral a unit absorbs, so used-capital clearing is automatic.

E. Aggregation and Stationary Equilibrium

Let $G(a, z, y)$ denote the stationary distribution of agents. Occupation is a function of the current state and so is not a separate coordinate of G . Define

$$\mathcal{E} = \{(a, z, y) : V_e(a, z, y) \geq V_w(a, z, y)\}, \quad (23)$$

$$\mathcal{W} = \{(a, z, y) : V_e(a, z, y) < V_w(a, z, y)\}. \quad (24)$$

Aggregate input demands and labor supply are

$$K_n^d = \int_{\mathcal{E}} K_n(a, z) dG, \quad K_u^d = \int_{\mathcal{E}} K_u(a, z) dG, \quad (25)$$

$$L^d = \int_{\mathcal{E}} L(a, z) dG, \quad L^s = \int_{\mathcal{W}} y dG. \quad (26)$$

Total deposits are $D = \int a dG$. By the balance-sheet condition (21), deposits equal the market value of the capital the intermediary holds. Combining this with the supply ratio (22) gives both stocks as functions of aggregate wealth:

$$K_n^s = \frac{D}{p_n + p_u \zeta / (\delta + \kappa)}, \quad K_u^s = \frac{\zeta}{\delta + \kappa} K_n^s. \quad (27)$$

Aggregate external finance is the value of operated capital in excess of own net worth,

$$B^{\text{ext}} = \int_{\mathcal{E}} \max \{0, p_n K_n(a, z) + p_u K_u(a, z) - a\} dG, \quad (28)$$

and its ratio to output is the leverage moment that disciplines θ . Markets clear when

$$K_n^d = K_n^s, \quad K_u^d = K_u^s, \quad L^d = L^s. \quad (29)$$

Given (27), these three conditions are equivalent to clearing the labor market, clearing the new-capital, and requiring aggregate demand to reproduce the technological ratio $K_n^d/K_u^d = (\delta + \kappa)/\zeta$. Used-capital clearing is then automatic.

Definition. A stationary competitive equilibrium consists of prices (W, r, R_n) , the implied capital prices $p_u = p_n - (R_n - r - \delta)/\zeta$ and $R_u = r + \delta + \kappa$, value functions $\{V_e, V_w, \Omega\}$, policy functions, firm input demands, capital stocks (K_n^s, K_u^s) , and a distribution G such that:

1. given (W, R_n, R_u, p_u) , firm input demands solve (8)–(9);
2. given prices and the continuation value Ω , consumption, saving, and occupational policies solve (10)–(15);
3. the intermediary's investment policy is optimal given (r, R_n) , which by Proposition 1 requires (19)–(18) and delivers zero profit and the balance sheet (21), with the stocks evolving according to (16)–(17);
4. G is invariant under the optimal saving and occupational policies and the transition processes (5) and (7);
5. the labor, new-capital, and used-capital clear as in (29).

F. Used Capital and Financial Constraints

The capital mix. Let $\mu(a, z) \geq 0$ be the multiplier on (9). For an interior choice of both new and used capital, the firm's first-order conditions are

$$\text{MPK}_n = p_n(R_n + \mu), \quad \text{MPK}_u = p_u(R_u + \mu). \quad (30)$$

Both capital types are used in positive amounts at any finite shadow prices. Although $\gamma > 1$ makes the two types gross substitutes, the marginal product of each remains unbounded at zero: as $K_n \rightarrow 0$ with $K_u > 0$ fixed, effective capital converges to $(1 - \phi)^{1/(\gamma-1)} K_u > 0$, so $\partial K / \partial K_n = \phi^{1/\gamma} (K/K_n)^{1/\gamma} \rightarrow \infty$, and symmetrically for K_u . Both conditions in (30) therefore hold with equality at any finite shadow prices. The two capital types are treated symmetrically. The firm pays $p_j R_j$ to rent one unit of type j and bears the additional shadow cost $p_j \mu$ for the financing capacity that the unit absorbs. Neither type receives a preferential collateral coefficient; they differ only in the market value p_j at which they enter (9). Combining (30) with (4) gives the optimal capital mix $\rho \equiv K_u / K_n$:

$$\rho(\mu) = \frac{1 - \phi}{\phi} \left[\frac{p_n(R_n + \mu)}{p_u(R_u + \mu)} \right]^\gamma. \quad (31)$$

At $\mu = 0$, this becomes $\rho^{uc} \equiv \rho(0)$. Because it depends only on market prices, all unconstrained firms choose the same mix of new and used capital, regardless of their wealth or productivity. The mix varies across firms only through the multiplier μ .

Collateral productivity. How a firm splits its capital between new and used machines also determines the scale that a given amount of financing capacity can support. Define

$$\Lambda(\rho) \equiv \frac{K}{p_n K_n + p_u K_u} = \frac{\Psi(\rho)}{p_n + p_u \rho}, \quad \Psi(\rho) \equiv \left[\phi^{1/\gamma} + (1 - \phi)^{1/\gamma} \rho^{(\gamma-1)/\gamma} \right]^{\gamma/(\gamma-1)}. \quad (32)$$

The function $\Lambda(\rho)$ measures the units of effective capital obtained per dollar of financing capacity. Define $\Lambda^{uc} \equiv \Lambda(\rho^{uc})$. A firm with a binding constraint exhausts (9) and therefore operates $\Lambda(\rho(\mu))\theta a$ units of effective capital. Appendix B derives the corresponding policies in closed form conditional on the multiplier, which is itself the root of a scalar equation. A constrained firm therefore has two adjustment margins: its scale and its choice of new versus used capital, which determines how far a given collateral position stretches.

Proposition 2 (Used-capital tilt and collateral relief). *Suppose*

$$R_u > R_n, \quad (33)$$

or, equivalently, $\kappa > \zeta(p_n - p_u)$. A sufficient condition is $\zeta \leq \kappa$, which by the stationary aggregate-stock relation (22) can equivalently be written as

$$\frac{K_u^s}{K_n^s} \leq \frac{\kappa}{\delta + \kappa}.$$

Then:

- (i) $\rho'(\mu) > 0$: *the used-to-new capital ratio increases strictly and monotonically from ρ^{uc} toward*

$$\bar{\rho} = \frac{1 - \phi}{\phi} \left(\frac{p_n}{p_u} \right)^\gamma,$$

the mix that minimizes the collateral value $p_n K_n + p_u K_u$ required per unit of effective capital.

- (ii) *Collateral productivity $\Lambda(\rho(\mu))$ is strictly increasing in μ on $[0, \infty)$ and converges to*

$$\bar{\Lambda} = \left[\phi p_n^{1-\gamma} + (1 - \phi) p_u^{1-\gamma} \right]^{1/(\gamma-1)}. \quad (34)$$

Moreover, $\bar{\Lambda} > p_n^{-1}$ whenever $p_u < p_n$ and $\gamma > 1$.

The proof is in Appendix B. Proposition 2 implies that constrained firms choose a larger used share than otherwise identical unconstrained firms, and that this share rises with the tightness of the constraint. This is the model counterpart of the firm-level gradients in Section III.⁵ The tilt does not require used capital to be cheaper to rent. Condition (33) is compatible with $p_u R_u > p_n R_n$: what matters is the financing capacity a unit absorbs per dollar of rental spending, not the rental payment itself.⁶ The relief is nevertheless bounded: because $\Lambda(\rho(\mu))$ converges to the finite limit in (34), tilting toward used capital economizes on financing capacity but cannot fully substitute for scale.⁷

Two margins. Used capital therefore affects both firm scale and occupational selection. Along the intensive margin, it allows constrained entrepreneurs to operate closer

⁵Condition (33) holds at the parameters calibrated in Section V.

⁶Eisfeldt and Rampini (2007) also predict that constrained firms use more used capital, through the timing of cash flows relative to purchase cost rather than through collateral capacity.

⁷The mechanism belongs to the market-value collateral framework of Rampini and Viswanathan (2013) and is closest to Rampini (2019), where an asset's durability determines the down payment it requires. Here durability varies *within* a single asset class that firms combine, so the response is a change in the mix rather than a move between asset classes.

to their unconstrained scale, reducing the dispersion of marginal revenue products. Along the extensive margin, it raises the value of entrepreneurship for productive but poor potential entrants and shifts the occupational boundary (15) accordingly.

V. Quantitative Analysis

Assigned parameters. Seven parameters are assigned outside the calibration. Following the literature, I set the labor share $\alpha = 2/3$, annual depreciation $\delta = 0.06$, the span of control $\eta = 0.79$, and relative risk aversion $\sigma = 1.5$, all adopted from Buera et al. (2011), and the annual persistence of worker ability $\rho_y = 0.95$, from Cagetti and De Nardi (2006). The remaining two govern the new-and-used capital aggregate (4). I set the CES weight on new capital to $\phi = 0.70$. This weight is pinned rather than estimated, because it is only weakly identified. The aggregate new-to-used ratio is a market-clearing condition that ζ delivers at any value of ϕ , so ϕ reaches the targeted moments only through the used-share gradient, and it moves that moment slowly: Appendix D raises ϕ from 0.70 to 0.75, the weight used by Lanteri (2018) for a second-hand capital, leaving the headline results unchanged. I set the elasticity of substitution to $\gamma = 2$. Edgerton (2011) estimates this elasticity from relative-price regressions derived from a CES composite of new and used equipment, and reports values between 1.7 and 2.0 for farm machinery and between 1.9 and 2.4 for construction machinery; $\gamma = 2$ lies in the overlap.

Jointly calibrated parameters. Seven parameters are calibrated jointly in the benchmark: $\{\theta, \eta_p, \omega, \sigma_y, \beta, \kappa, \zeta\}$. Each is disciplined primarily by one moment, though all seven are selected together. The debt-to-output ratio of 0.600, computed as B^{ext}/Y from (28), disciplines the collateral multiplier θ . The employment share of the largest decile of firms, 0.632, disciplines the Pareto-tail parameter η_p of entrepreneurial productivity. The annual firm exit rate of 0.098, which I compute from the survey and which is consistent with the 9–10 percent annual death rate reported by Berkel et al. (2020), disciplines productivity persistence ω . Agents in the model are infinitely lived and firms do not die, so the moment is mapped to the rate at which entrepreneurs switch to wage work; the survey records the closure or suspension of the household’s enterprise rather than the disappearance of the household, which is what that switch is here. The worker wage Gini of 0.372, reported by Doan et al. (2023) from the Vietnam Household Living Standard Survey, disciplines the innovation dispersion σ_y of worker ability. The annual real interest rate of 0.031, the Vietnamese deposit rate net of GDP-deflator inflation (World Bank, 2026), is matched by the discount factor: the model requires $\beta = 0.858$ to clear the deposit market at that rate.

Used-capital parameters. Two moments jointly determine κ and ζ . The first is the aggregate ratio of new to used capital, which in a stationary equilibrium is

$$\frac{K_n^s}{K_u^s} = \frac{\delta + \kappa}{\zeta}, \quad (35)$$

and captures the relative *quantity* of the two capital types in the economy. The second captures the distribution of new and used capital across firms using the cross-sectional relationship between the used-capital share and firm size:

$$s_{u,i} = b_0 + b_1 \log A_i + \varepsilon_i, \quad b_1 = -0.027 \text{ (0.0061)}, \quad N = 886. \quad (36)$$

I estimate this relationship by ordinary least squares using the 2013 cross-section of firms reporting a positive used-capital share. The dependent variable $s_{u,i}$ is the used-capital share on a unit interval, so $b_1 = -0.027$ means that a one-log-point increase in assets is associated with a 2.7 percentage-point decline in the used share, or equivalently -0.027 on the 0–1 scale. This is a *between-firm* size gradient in one cross-section. It is the natural target for a stationary model and plays a different role from the within-firm zero-debt interactions in Table 2, which identify off firms whose debt status and assets change over the panel. Whereas the first moment captures the relative quantity of new and used capital, the second primarily reflects their relative price in collateral terms. If used capital is cheaper, smaller firms, which are more likely to face binding collateral constraints, rely more heavily on it, making b_1 more negative. Conversely, a smaller price advantage for used capital produces a flatter size gradient. The two moments imply $\kappa = 0.143$ and $\zeta = 0.079$. At these values, the condition of Proposition 2 is satisfied: $R_u = r + \delta + \kappa = 0.234$ exceeds $R_n = r + \delta + \zeta(p_n - p_u) = 0.119$, and the sufficient condition $\zeta \leq \kappa$ holds as well.

Comparison design. I compare two economies to isolate the role of the used-capital: a benchmark in which firms rent both new and used capital, and an economy without used capital. Both are calibrated to the same five common moments; the benchmark additionally matches the used-capital-share gradient and the aggregate new-to-used ratio. Removing used capital ($\zeta = 0$, $\phi = 1$) leaves the five common moments to be matched by recalibrating $\{\theta, \eta_p, \omega, \sigma_y, \beta\}$, and I solve for the general-equilibrium prices and allocations separately in each economy. Because both economies attain the same fit to the common targets, the differences between them reflect the absence of the used-capital. The lower value of θ is the value required to reproduce the benchmark’s debt-to-output ratio.

Table 3. Calibration: Parameters, Values, and Targets

Parameter	Benchmark	Model without used capital	Source or target
<i>Panel A: Parameters assigned from the literature</i>			
α : Labor share	0.667	0.667	Buera et al. (2011)
δ : Depreciation rate	0.060	0.060	Buera et al. (2011)
η : Span of control	0.790	0.790	Buera et al. (2011)
σ : Risk aversion	1.500	1.500	Buera et al. (2011)
ρ_y : Ability persistence	0.950	0.950	Cagetti and De Nardi (2006)
ϕ : CES weight on new capital	0.700	1.000	Assigned; see text
γ : Substitution, new vs. used	2.000	—	Edgerton (2011)
<i>Panel B: Parameters jointly calibrated to data moments</i>			
θ : Collateral multiplier	2.143	1.985	Debt-to-output ratio: 0.600
η_p : Productivity-tail parameter	5.949	5.397	Top-decile employment share: 0.632
ω : Productivity persistence	0.861	0.836	Firm exit rate: 0.098
σ_y : Ability innovation dispersion	0.245	0.239	Worker wage Gini: 0.372
β : Discount factor	0.858	0.854	Real interest rate: 0.031
ζ : New-to-used transition rate	0.079	0.000	New-to-used capital ratio: 2.570
κ : Breakage rate of used capital	0.143	—	Used-share gradient: -0.027

Notes: Most targets are computed on the Vietnamese Small and Medium Enterprise Survey (CIEM et al., 2005–2015). The two economies are calibrated separately, as described below.

A. The Mechanism of Used Capital

Used capital works through the collateral constraint. A firm that cannot meet the constraint operates below its efficient scale. The multiplier μ measures the shadow cost of the missing financing capacity. The left panel of Figure 1 plots this shadow cost by benchmark entrepreneur wealth decile, separately for low, mid, and high entrepreneurial productivity z . It is highest for entrepreneurs who combine high productivity with little wealth—precisely the firms for which the constraint binds most strongly—and declines with wealth but remains positive for most entrepreneurs in the stationary distribution as wealth accumulates.

The right panel maps the shadow cost into the capital mix. An unconstrained firm faces $\mu = 0$ and chooses the mix of new and used capital that minimizes its rental cost at prevailing prices. Constrained firms depart from this benchmark. A dollar of rental spending on capital type j purchases $1/(p_j R_j)$ units, each of which absorbs p_j units of financing capacity. Consequently, a dollar of rental spending on type j uses $1/R_j$ units of financing capacity. Because $R_u > R_n$, used capital absorbs less financing capacity per dollar of rental spending than new capital. The used-capital share therefore rises monotonically with μ and approaches the ceiling implied by $\bar{\rho}$ in Proposition 2.

Taken together, the two panels show that the firms facing the tightest collateral constraints are precisely those that tilt most strongly toward used equipment. This tilt is a response to financing constraints rather than a difference in production technology. By reallocating financing capacity toward used equipment, constrained firms can

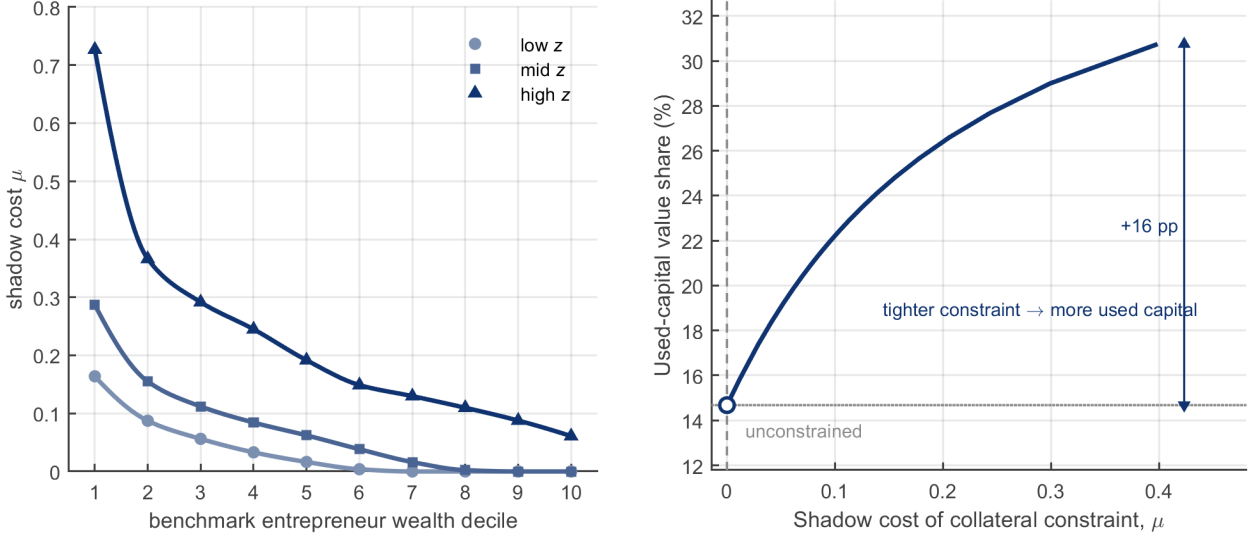


Figure 1. The mechanism of used capital. Left panel: the shadow cost of the collateral constraint, μ , by benchmark entrepreneur wealth decile, for low, mid, and high entrepreneurial productivity z . Right panel: the used-capital value share, $100 p_u K_u / (p_n K_n + p_u K_u)$, as a function of μ at benchmark equilibrium prices. The horizontal axis covers the range containing most constrained firms, and the open circle marks the unconstrained firm at $\mu = 0$.

obtain more effective capital from a given collateral position. This mechanism drives the allocation and entry results that follow.

B. Used Capital and Capital Misallocation

I measure the cost of financial frictions by comparing each calibrated economy with its own frictionless counterpart. In each case I set θ large enough that the collateral constraint is slack for every operating firm. I hold all other parameters fixed and recompute equilibrium prices and the stationary distribution. No agent with strictly positive wealth is then constrained, and MRPK dispersion is numerically zero. Using a separate frictionless counterpart for each economy isolates the misallocation caused by financial frictions from differences in technology, calibration, and equilibrium prices.

Define the TFP loss from the collateral constraint as

$$\mathcal{M} = 100 \log \left(\frac{\text{TFP}^{\text{frictionless}}}{\text{TFP}^{\text{friction}}} \right), \quad (37)$$

measured in log points against the economy's own frictionless counterpart.

Relative to its own frictionless counterpart, the collateral constraint generates a TFP loss of $\mathcal{M} = 5.72$ log points in the benchmark. The same constraint generates $\mathcal{M} = 9.39$ log points in the model without used capital, recalibrated to the same

targets. Second-hand capital therefore offsets 3.67 log points, or

$$\frac{9.39 - 5.72}{9.39} = 39.1\%$$

of the frictional loss that arises in its absence. This decomposes the model’s misallocation into the portion the used-capital margin removes. Table 4 reports this comparison together with the allocation statistics behind it.

Table 4. Used Capital and Capital Misallocation

	Benchmark	No used capital	Effect of used capital
TFP loss, $\mathcal{M}$ (log points)	5.72	9.39	39.1% eliminated
Mean scale relative to target, K/K^*	0.343	0.279	23.0% higher
sd(log MRPK)	0.450	0.598	24.7% lower
Share of firms constrained	0.908	0.909	0.1% lower

Notes: The benchmark contains new and used capital, while the other model is without used capital and recalibrated to the common targets. The TFP loss $\mathcal{M}$ is defined in (37) and is computed within each economy against its own frictionless counterpart, so the two columns are not measured against a common benchmark. K/K^* compares each firm’s new capital to the amount it would install if the collateral constraint did not bind at the economy’s prices; the table reports the average across firms (very high ratios are capped so a few outliers do not dominate). A value of one means firms operate at that unconstrained scale on average; lower values mean they fall short. The sd(log MRPK) row measures how much marginal products differ across firms: a higher number means more dispersion and more misallocation. Percentages in the last column compare the benchmark with the model without used capital and are computed from the unrounded values.

Second-hand capital reallocates capital toward financially constrained, high-return firms. Access to used capital allows constrained entrepreneurs to operate closer to their unconstrained scale, as Figure 2 shows. The effect is largest among low-wealth entrepreneurs. Second-hand capital therefore directs financing capacity toward firms with the highest marginal returns to capital.

The single-vintage economy is separately calibrated. The headline estimate therefore also incorporates changes in the fitted productivity distribution. Holding that distribution fixed instead gives 19.5 percent. Appendix C reports all three comparison designs. No margin changes sign across them.

C. Used Capital and Entry by Productive Poor Agents

The second main result concerns occupational selection. I define “talented” agents as those in the top decile of entrepreneurial productivity z . “Poor” agents are those with wealth below the benchmark median. Access to used capital raises entrepreneurship among poor talented agents from 34.3 to 42.0 percent, an increase of 7.7 percentage points. By contrast, entrepreneurship among rich talented agents is nearly unchanged.

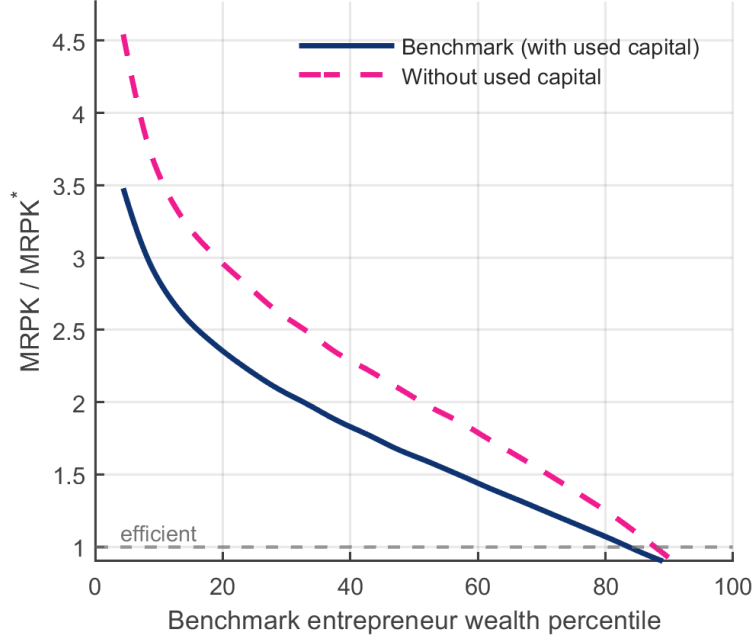


Figure 2. Used capital and the allocation of capital. Each point is a firm’s marginal product of capital (MRPK) divided by the average MRPK in that economy’s frictionless counterpart (MRPK*), plotted by benchmark wealth percentile. Productivity is held fixed within each wealth bin, so the profile reflects financial constraints rather than sorting by ability. Solid line: benchmark with used capital; dashed line: model without used capital. MRPK* is an economy-wide average in the frictionless economy, not the MRPK of an unconstrained firm; a ratio near one therefore does not mean the collateral constraint is slack ($\mu = 0$).

Used capital therefore reduces the dependence of entrepreneurship on initial wealth rather than increasing entry uniformly. Table 5 reports these entry rates together with their frictionless counterparts.

Table 5. Used Capital and Entry by Productive Poor Agents

	Benchmark	No used capital	Effect of used capital
Poor talented entrepreneurship rate	0.420	0.343	+7.7 p.p.
Rich talented entrepreneurship rate	0.827	0.833	−0.6 p.p.
Poor talented rate, frictionless	0.714	0.682	—
Rich talented rate, frictionless	0.801	0.744	—
Entry shortfall, $\mathcal{S}$ (log points)	53.2	68.7	22.6% repaired

Notes: “Talented” denotes the top productivity decile and “poor” denotes wealth below the benchmark population median. These cutoffs are constructed once in the benchmark and applied in levels to both economies. Each economy is compared with its own frictionless counterpart, so the frictionless rows differ across columns. The entry shortfall $\mathcal{S}$ in (38) is defined for poor talented agents.

The mechanism operates through the occupational boundary. Used capital absorbs less financing capacity. Productive agents with little wealth can therefore operate at a larger scale. This raises profits $\pi(a, z)$ and the value of entrepreneurship $V_e(a, z, y)$, enabling some of them to enter. Rich talented agents already possess sufficient financing capacity. Here “rich” means above the population median; the constraint

may still bind at the margin, but used capital changes their scale and entry little relative to poor talented agents. Their occupational choices therefore change little.

To account for poor talented agents who prefer wage work even without financial frictions, I define the collateral-induced entry shortfall as

$$\mathcal{S} = 100 \log \left(\frac{e^{\text{frictionless}}}{e^{\text{friction}}} \right), \quad (38)$$

where e is their entrepreneurship rate. The shortfall is $\mathcal{S} = 53.2$ log points in the benchmark and $\mathcal{S} = 68.7$ log points in the model without used capital. Thus, used capital repairs

$$1 - \frac{53.2}{68.7} = 22.6\%$$

of the entry distortion caused by the collateral constraint.

For rich talented agents, the shortfall is negative in both economies. Entry under the constraint (0.827 in the benchmark) exceeds entry in its own frictionless counterpart (0.801), implying a shortfall of -3.2 log points. The corresponding figure without used capital is -11.3 log points. This is a general-equilibrium effect rather than a relaxation of the constraint. Removing the collateral constraint raises labor demand and hence the wage. This raises the value of wage work and draws some marginal high-productivity agents out of entrepreneurship. The effect is confined to agents whose financing capacity was never binding. More precisely, the wage effect operates on high productivity agents who are unconstrained relative to the occupational margin—those who enter even under frictions—not on agents with $\mu = 0$.

VI. Conclusion

This paper begins from a discrepancy between standard models and the data. In models with homogeneous capital, a financially constrained firm can respond to limited credit only by operating at a smaller scale. In the data, however, young, small, and credit-excluded firms also change the mix of their capital. They rely more heavily on second-hand equipment. This paper studies whether that additional adjustment margin changes the allocation and occupational consequences of financial frictions. The mechanism follows from the way collateral constraints are defined. The constraint limits the market value of the capital a firm operates. Used capital is more expensive to operate, because it depreciates faster. But its market price is lower. A dollar spent renting used equipment therefore ties up less collateral than a dollar spent renting new. A productive entrepreneur with little wealth can use second-hand equipment to operate more capital than new equipment alone would allow. As the entrepreneur

accumulates wealth, the constraint becomes less severe. The advantage then declines, and the firm shifts toward the unconstrained mix. This mechanism accounts for the concentration of used equipment among young, asset-poor, and credit-excluded firms in the Vietnamese data. Quantitatively, access to second-hand capital reduces the misallocation that the collateral constraint generates by 39 percent. The comparison is with a single-vintage economy in which firms rent only new capital. It is calibrated on its own to the same five empirical targets; because the vintage margin is absent, its calibrated parameters differ from the benchmark's. Used equipment allows productive firms with little wealth to operate closer to their efficient scale, and it reduces the dispersion of marginal products across firms. The same mechanism affects occupational choice. Without access to used capital, productive but poor agents are less likely to become entrepreneurs. Entry among productive and wealthy agents changes little.

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Appendix A. The Intermediary’s Problem

Proof of Proposition 1. The dynamic problem of the intermediary is

$$V(K_n, K_u) = \max_{K'_n, I} \left\{ R_n K_n + p_u R_u K_u - I + \frac{V(K'_n, K'_u)}{1 + r} \right\}, \quad (\text{A1})$$

subject to

$$K'_n = (1 - \delta - \zeta)K_n + I, \quad K'_u = (1 - \delta - \kappa)K_u + \zeta K_n.$$

Take the first-order derivatives with respect to K_n and K_u :

$$\frac{\partial V(K_n, K_u)}{\partial K_n} = R_n + (1 - \delta - \zeta) + \frac{\partial V(K'_n, K'_u)}{\partial K'_u} \frac{\zeta}{1 + r}, \quad (\text{A2})$$

$$\frac{\partial V(K_n, K_u)}{\partial K_u} = p_u R_u + \frac{\partial V(K'_n, K'_u)}{\partial K'_u} \frac{1 - \delta - \kappa}{1 + r}. \quad (\text{A3})$$

The first-order condition with respect to K'_n is

$$\frac{\partial V(K_n, K_u)}{\partial K'_n} = -1 + \frac{\partial V(K'_n, K'_u)}{\partial K'_n} \frac{1}{1 + r} = 0. \quad (\text{A4})$$

Equation (A3) implies

$$\frac{\partial V(K_n, K_u)}{\partial K_u} = p_u R_u + \frac{1 - \delta - \kappa}{1 + r} p'_u R'_u + \left(\frac{1 - \delta - \kappa}{1 + r} \right)^2 p''_u R''_u + \dots$$

At the steady state this is equivalent to

$$\frac{\partial V(K_n, K_u)}{\partial K_u} = p_u R_u \frac{1 + r}{r + \delta + \kappa}. \quad (\text{A5})$$

Combine (A2) and (A4) to obtain

$$1 + r = R_n + (1 - \delta - \zeta) + \frac{\zeta}{1 + r} \frac{\partial V(K''_n, K''_u)}{\partial K''_u}. \quad (\text{A6})$$

Applying the steady-state condition (A5),

$$r = R_n - \delta - \zeta \left(1 - \frac{p_u R_u}{r + \delta + \kappa} \right), \quad (\text{A7})$$

which is (18). □

The split into a price and a user cost. Only the product $p_u R_u$ appears in (A7), so something more is needed to say what each factor is. Let p_u be the price a used unit actually trades at, the same price that is carried on the balance sheet (21) and in the financing constraint (9). Buy a unit today for p_u . Over the period it pays the rental $p_u R_u$, and at the end of the period what is left of it is worth $(1 - \delta - \kappa)p_u$. The

two together have to be worth $(1+r)p_u$, or no one would hold it:

$$(1+r)p_u = p_u R_u + (1-\delta-\kappa)p_u.$$

This gives $R_u = r + \delta + \kappa$, and (A5) then reads $\partial V / \partial K_u = (1+r)p_u$. So p_u is what a used unit is worth today with this period's rental still to come; once that rental has been paid, what is left is the smaller amount $(1-\delta-\kappa)p_u$. Read the same line as $p_u R_u = r p_u + (\delta + \kappa) p_u$: the rental has to cover interest on what the unit is worth, its depreciation, and the chance that it breaks. That is why R_u is a user cost. Equation (A7) now reads $R_n = r + \delta + \zeta(p_n - p_u)$, and inverting it delivers (20). Only the product enters any firm's rental bill. The financing constraint (9), written on market value, is what makes the two factors matter separately.

Zero profit. The technology is constant returns and (A1) is linear, so V is homogeneous of degree one. By Euler's theorem, $V = V_n K_n^s + V_u K_u^s = (1+r)(p_n K_n^s + p_u K_u^s)$. Deposits are the only liability and capital the only asset, which is (21). The intermediary therefore owes $(1+r)D$ against a portfolio worth V , and profit is zero. This establishes the claim made in Section IV.D.

Appendix B. Derivations

A. The capital mix

Proof of Proposition 2(i). A physical unit of type j costs $p_j R_j$ to rent and absorbs p_j of financing capacity. At shadow cost $\mu \geq 0$ its effective price is therefore $p_j(R_j + \mu)$, and for a given level K of effective capital the entrepreneur solves

$$\min_{K_n, K_u} p_n(R_n + \mu)K_n + p_u(R_u + \mu)K_u \quad \text{s.t.} \quad \left[\phi^{1/\gamma} K_n^{\frac{\gamma-1}{\gamma}} + (1-\phi)^{1/\gamma} K_u^{\frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}} = K. \quad (\text{B1})$$

The first-order conditions give the CES demand ratio (31), which factors as

$$\rho(\mu) = \bar{\rho} \left[\frac{R_n + \mu}{R_u + \mu} \right]^\gamma, \quad \bar{\rho} \equiv \frac{1-\phi}{\phi} \left(\frac{p_n}{p_u} \right)^\gamma, \quad (\text{B2})$$

since the price ratio does not depend on μ . In particular $\rho^{uc} = \bar{\rho} (R_n/R_u)^\gamma$. Differentiating the bracket,

$$\frac{\partial}{\partial \mu} \frac{R_n + \mu}{R_u + \mu} = \frac{R_u - R_n}{(R_u + \mu)^2}, \quad (\text{B3})$$

so $\rho'(\mu) > 0$ if and only if $R_u > R_n$, which is (33). As $\mu \rightarrow \infty$ the bracket tends to one. Hence $\rho(\mu) \rightarrow \bar{\rho}$. $\square$

B. Collateral productivity

Proof of Proposition 2(ii). Normalize $K_n = 1$, so that $\rho = K_u$ and the bundle is $\Psi(\rho) = [\phi^{1/\gamma} + (1 - \phi)^{1/\gamma} \rho^{\frac{\gamma-1}{\gamma}}]^{\frac{\gamma}{\gamma-1}}$. The capacity absorbed is $p_n + p_u \rho$, so collateral productivity is $\Lambda(\rho) = \Psi(\rho)/(p_n + p_u \rho)$, as in (32). Maximizing Λ is equivalent to minimizing the collateral cost of one unit of the bundle, $(p_n + p_u \rho)/\Psi(\rho)$. That function is strictly quasi-convex in ρ : for every $t > 0$ the map $\rho \mapsto p_n + p_u \rho - t\Psi(\rho)$ is strictly convex, since Ψ is strictly concave, so each sublevel set is an interval. Its minimizer is therefore unique, and the first-order condition returns the $\bar{\rho}$ of (B2): the mix that minimizes collateral at prices (p_n, p_u) is the same mix that (B2) approaches as $\mu \rightarrow \infty$, because the user costs drop out of the demand ratio in that limit. Hence

$$\bar{\Lambda} \equiv \Lambda(\bar{\rho}) = \left[\phi p_n^{1-\gamma} + (1 - \phi) p_u^{1-\gamma} \right]^{1/(\gamma-1)}, \quad (\text{B4})$$

which is (34), and Λ is strictly increasing on $(0, \bar{\rho})$.

By (B2), $\rho(\mu)$ is $\bar{\rho}$ times a bracket that under (33) rises to one from below. Therefore $\rho(\mu)$ increases from $\rho^{uc} < \bar{\rho}$ toward $\bar{\rho}$, and $\Lambda(\rho(\mu))$ is strictly increasing in μ with limit $\bar{\Lambda}$. Finally $\bar{\Lambda} > p_n^{-1}$ whenever $p_u < p_n$ and $\gamma > 1$, since then $p_u^{1-\gamma} > p_n^{1-\gamma}$ and the convex combination in (B4) exceeds $p_n^{1-\gamma}$. $\square$

Two benchmarks should be distinguished. The p_n^{-1} of Proposition 2(ii) is collateral productivity in the economy without used capital, where $K = K_n$ and a dollar of collateral supports $1/p_n$ units of the bundle; it is the counterpart of the recalibrated no-used economy of Section V. It is not $\Lambda(0) = \phi^{1/(\gamma-1)}/p_n$, the productivity of a firm that retains the share parameter $\phi < 1$ but is denied used capital. Since $\phi < 1$ and $\gamma > 1$, $\Lambda(0) < p_n^{-1}$, so the ratio $\bar{\Lambda}/\Lambda(0)$ combines the financing gain with a mechanical penalty from a missing CES input; $\bar{\Lambda} > \Lambda(0)$ holds by optimality alone and requires no condition on prices.

C. Which margin the relief runs on

Three price comparisons should be kept apart. Used capital is cheaper per physical unit, $p_u < p_n$, which by (20) is equivalent to $R_n > r + \delta$; this is what delivers $\bar{\Lambda} > p_n^{-1}$, and prices are the whole of that result. Its user cost per dollar of market value is higher, $R_u > R_n$ under (33); this is what delivers the tilt. Its rental payment per physical unit, $p_u R_u$, may lie on either side of $p_n R_n$, and no result here depends on which. Both $p_u < p_n$ and $R_u > R_n$ are equilibrium properties rather than restrictions on primitives, and both hold at the calibrated values.

The effective price $p_j(R_j + \mu)$ in (B1) already encodes the trade-off, and reading it

per rental dollar makes the mechanism plain. A dollar of rental spending on type j hires $1/(p_j R_j)$ units and ties up

$$\frac{p_j}{p_j R_j} = \frac{1}{R_j} \quad (\text{B5})$$

of financing capacity. The price cancels, so on this margin the user cost alone decides: renting used capital frees collateral relative to new exactly when $R_u > R_n$, the equivalent forms of which are given in Proposition 2. What the constraint counts is market value, and a rental dollar spent on used capital commands less of it.

Appendix C. The Comparison Design and the Productivity Tail

The comparison in the text recalibrates the economy without used capital in full. Its Pareto-tail parameter therefore falls from 5.949 to 5.397. A fatter tail places more mass on highly productive entrepreneurs, who have the largest desired scale, and so raises measured misallocation on its own. This appendix holds the tail fixed in two ways. The *tail-pinned* economy keeps $(\eta_p, \omega, \sigma_y)$ at their benchmark values and recalibrates only (θ, β) . The *tail diagnostic* returns η_p to 5.949 in the fully recalibrated economy and re-clears prices without refitting any moment. Its targeted moments therefore no longer hold exactly. Misallocation reduced, $1 - \mathcal{M}_{\text{bm}}/\mathcal{M}$, is the share of that design’s TFP loss $\mathcal{M}$ —defined in (37)—that the benchmark reduces. Entry repaired is $1 - \mathcal{S}_{\text{bm}}/\mathcal{S}$, defined the same way from the entry shortfall $\mathcal{S}$ in (38). Each economy is compared with its own frictionless twin. Changing η_p alone moves the TFP

Table 6. The result under three designs without used capital

Economy	η_p	TFP loss $\mathcal{M}$ (log pts)	Misallocation reduced	Entry repaired	K/K^*	sd(logMRPK)
Benchmark (used K)	5.949	5.72	—	—	0.343	0.4501
No used capital, tail pinned	5.949	7.11	19.5%	32.0%	0.268	0.5298
Tail diagnostic	5.949	7.42	22.9%	27.8%	0.295	0.5296
No used capital, recalibrated	5.397	9.39	39.1%	22.6%	0.279	0.5977

loss from 7.42 to 9.39 log points. The collateral multiplier, productivity persistence, worker dispersion, the discount factor and equilibrium prices together account for 0.31. Under all three designs the benchmark has the lower TFP loss, the higher mean scale relative to target, the lower dispersion of log MRPK, and the higher rate of entry by poor talented agents. The share of the frictional loss attributed to used capital varies with the design. Its sign does not. The tail-pinned and tail-diagnostic economies report nearly identical sd(logMRPK) (0.5298 vs. 0.5296): both retain the benchmark productivity tail and remain in the same high-dispersion regime without used capital, so restoring η_p in the diagnostic moves mean scale (K/K^* from 0.268 to

0.295) more than it moves cross-firm MRPK dispersion. The two margins move in opposite directions. Misallocation reduced rises from 19.5 to 39.1 percent as the tail is refit, while entry repaired falls from 32.0 to 22.6. A fatter tail places more mass on very productive entrepreneurs, whose desired scale is largest, so constraining them is expensive and measured misallocation rises. But those same agents are inframarginal in the occupational choice (productive enough to enter even at a constrained scale, so the constraint blocks fewer of them and the entry shortfall is smaller.)

Appendix D. Robustness to the Assigned CES Parameters

The weight ϕ on new capital and the elasticity of substitution γ in the capital aggregator (4) are assigned rather than calibrated; the benchmark sets $\phi = 0.70$ and $\gamma = 2.0$. Table 7 re-solves the benchmark at $\phi = 0.75$, the value used by Lanteri (2018), and at $\gamma \in \{1.7, 2.0, 2.5\}$, which brackets the range Edgerton (2011) estimates, in each case holding the calibrated parameters $(\theta, \eta_p, \omega, \kappa, \zeta, \sigma_y, \beta)$ at their benchmark values and re-clearing only the prices (W, r, R_n) . These are robustness sweeps rather than recalibrations: each asks whether the results turn on the assigned value, not whether the model could be re-fitted around it. The $\gamma = 2.0$ column reports the stored benchmark; the $\phi = 0.70$ column is the same economy and should match it. The other columns were solved in separate runs and should be read relative to $\gamma = 2.0$ or $\phi = 0.70$. Neither parameter matters for the conclusions: the TFP loss $\mathcal{M}$ moves by at most 0.06 log points, about one percent of its level, and misallocation reduced stays between 38.7 and 39.3 percent.

Table 7. Sensitivity of the results to the assigned CES parameters

	ϕ sweep		γ sweep		
	$\phi = 0.70$ (benchmark)	$\phi = 0.75$	$\gamma = 1.7$	$\gamma = 2.0$ (benchmark)	$\gamma = 2.5$
<i>Panel A: Gain from removing the collateral constraint</i>					
TFP (%)	5.89	5.91	5.92	5.89	5.86
TFP loss $\mathcal{M}$ (log points)	5.72	5.74	5.75	5.72	5.70
Misallocation reduced (%)	39.1	38.9	38.7	39.1	39.3
<i>Panel B: Allocation in the constrained equilibrium</i>					
Mean scale relative to target, K/K^*	0.343	0.350	0.347	0.343	0.337
Share of firms constrained	0.908	0.906	0.907	0.908	0.908
Entrepreneur share	0.221	0.220	0.221	0.221	0.220
<i>Panel C: Cleared prices</i>					
Wage W	0.540	0.542	0.540	0.540	0.540
Interest rate r	0.031	0.031	0.031	0.031	0.031
New-capital rental R_n	0.119	0.124	0.119	0.119	0.120
Used-capital price p_u	0.645	0.582	0.648	0.645	0.640

Notes: Each column re-clears (W, r, R_n) at the stated parameter value with the calibrated parameters held at their benchmark levels; the benchmark is $\phi = 0.70$, $\gamma = 2.0$. The $\phi = 0.70$ and $\gamma = 2.0$ columns refer to the same economy. $\mathcal{M}$ is the TFP loss from the collateral constraint defined in (37), measured against each economy's own frictionless counterpart; the TFP row is the same object in percent rather than log points.